Decomposing the GP Advantage in Offline MBO

Anonymous submission

Abstract

Offline model-based optimization ships a surrogate and an optimizer together and reports one number, leaving attribution between them open—as the subfield's own 2026 survey states (Kim et al. 2026). We cross both axes under one shared protocol—three surrogate classes against three search routines—and decompose the outcome variance between them. The crossed design returns the term a one-factor-at-a-time study cannot estimate, and it is our strongest result. On the synthetic grid the surrogate×optimizer interaction is significant: η^2_{inter} of 0.146–0.165 under min–max normalization, with a bootstrap interval excluding zero in all four engine corners and surviving bias correction (0.134–0.156). It exceeds the optimizer main effect in eleven of twelve corner×normalizer combinations, inverting only in the on/off corner under rank normalization, where the optimizer axis is most competitive. A design that fixes the acquisition rule and varies only the surrogate (Li, Rudner, and Wilson 2024) is structurally unable to estimate this term. It also resolves a real confusion: a small optimizer main effect (η^2_{opt}=0.038) licenses "optimizer choice explains little *marginal* variance" and never "optimizer choice is arbitrary," and the factorial measures the difference—the conditional effect is four to thirty-three times the marginal. On Design-Bench the interaction is smaller and subordinate (Section 6). Any η^2 headline is a joint artifact of five operating-point coordinates—ensemble size K, pessimism β, search budget, engine corner, and the per-task normalizer—so a headline that does not state all five is not reproducible. The dependence is measurable: the headline more than doubles across the search-budget axis (0.243 to 0.526, disjoint bootstrap intervals), two independent runs at one *fixed* operating point agree to 0.0018 while differing on 58 of 63 per-cell means, and leave-one-task-out moves it across 0.373–0.485. The instrument that exposes those coordinates is a taxonomy of five confounds specific to offline-MBO surrogate comparison, each traced to a line of the scoring path, each with a signed effect on the decomposition and a pre-registered kill criterion, of which the β–σ criterion fired. The confounds are the contribution and the crossed decomposition is the instrument: we find no prior offline-MBO work that reports a crossed surrogate×optimizer variance decomposition, and Policy-Guided Gradient Search (PGS)—whose Table 1 supplies our reference point—could not have produced one, since it fixes the surrogate as the operative factor and never crosses it (Chemingui et al. 2024). Correcting the two *protocol* confounds moves the surrogate effect *up* rather than down—from our decomposition of PGS's Table 1, $\eta^2_{\text{surr}} \approx 0.37$, to 0.405 $[0.290, 0.556]$, a direction we found no precedent for in the ML audit literature—while the corner intervals do not resolve at n=7 tasks, so we claim the direction, not a level. Across four corners and three normalizers $\eta^2_{\text{surr}} > \eta^2_{\text{opt}}$ in all twelve. Under a matched query budget the optimizer main effect in the synthetic grid is η^2_{opt}=0.038 $[0.003, 0.123]$, eight times its unmatched value—established at high budget, underpowered at low. On Design-Bench we detect no surrogate difference at this power, which at n=7 tasks is not equivalence; matching query budgets there raises the optimizer effect by 1.5 to $1.9\times$ in every corner rather than dissolving it. What the surrogate advantage *is* remains open, narrowed by seven controls (Section 5). That the statistical model often outranks the acquisition heuristic is a decade-old doctrine we do not contest (Shahriari et al. 2016); what this design falsifies is the local premise of one recent paper (Chemingui et al. 2024), not any field-level belief.

1 Introduction

Offline model-based optimization (MBO) seeks a design x maximizing an expensive black-box f from a fixed dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$, with no further queries to f during search (Trabucco et al. 2022; Kim et al. 2026). The dominant recipe fits a surrogate $\hat{f}$ and optimizes it; because optimization pushes designs into regions where $\hat{f}$ is unconstrained, the surrogate is typically made conservative—by a learned regularizer (Trabucco et al. 2021) or by a pessimistic acquisition such as the lower confidence bound (LCB) $\mu(x) - \beta\sigma(x)$ (Srinivas et al. 2012). A common intuition is that Gaussian-process LCB (GP-LCB) beats neural-ensemble LCB at low dimension because the GP posterior is better calibrated. But the two pipelines differ in *two* coupled factors, not one: the surrogate class (GP posterior vs. ensemble disagreement) and the acquisition optimizer, since GPs are usually optimized by perturbation or quasi-Newton restarts and neural surrogates by gradient ascent on x. Every method in this lineage ships both at once and reports one number.

The attribution question is open, by the field's own account. The subfield's 2026 survey states that existing benchmarks "often emphasize overall optimization performance without clarifying whether observed gains stem from

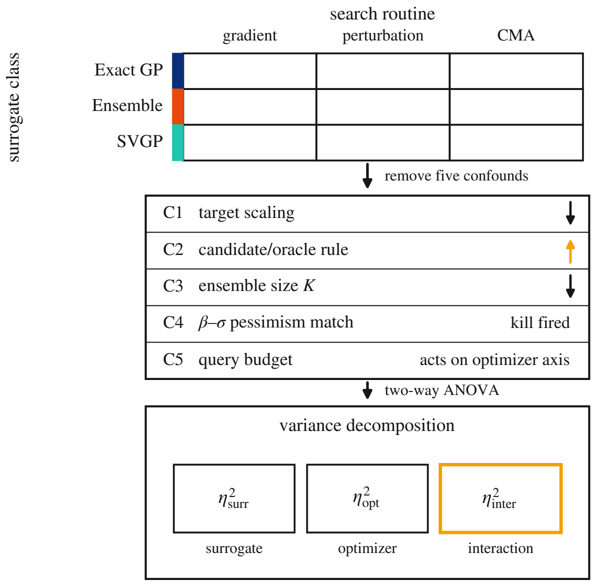

Figure 1: The design: a crossed surrogate×optimizer factorial—three surrogate classes against three search routines under one shared protocol—from which five confounds specific to offline-MBO surrogate comparison are removed (each traced to a line of the scoring path, each with a signed effect), then decomposed by two-way ANOVA into the surrogate and optimizer main effects and the surrogate×optimizer interaction the factorial exists to estimate.

superior surrogate modeling, improved optimization strategies, or mere chance" (Kim et al. 2026). That is not our characterization of a gap; it is the field certifying that the attribution is unresolved. Resolving it takes a design nobody has run: both axes manipulated against each other under one shared protocol, with the outcome variance decomposed between them. The systematic studies that exist fix one axis and vary the other, and the most thorough of them is *broader* than ours on the surrogate axis precisely because it fixes the other: a controlled seven-class Bayesian neural-network surrogate study that holds Monte-Carlo expected improvement fixed as its acquisition function for all problems, runs *online* sequential BO rather than offline MBO, never varies a search routine, and reports no variance decomposition (Li, Rudner, and Wilson 2024). A separate line varies the *training objective* inside a fixed architecture—swapping a ranking loss for regression inside one MLP (Tan et al. 2025), or adding a conservative regularizer (Trabucco et al. 2021)—which is a different kind of factor from model class, since it can be applied at fixed estimator family whereas a GP posterior cannot be instantiated without its marginal-likelihood fit. The adjacent offline-RL literature does the same: the closest re-evaluation there varies uncertainty-penalty heuristics while holding the policy optimizer fixed, flagging a differing optimizer only as an unresolved appendix confound (Lu et al. 2022). A design that fixes a factor cannot estimate that factor's effect, however many levels it gives the other—so a seven-class comparison at one fixed acquisition rule is not a smaller version of this experiment but a transposed one. This paper runs the crossed surrogate×optimizer factorial (Figure 1) and reports the two-way variance decomposition. The claim we make is about what the literature *reports*: no prior work in offline MBO reports a crossed surrogate-class × optimizer variance decomposition. We do not claim the design space forbade one—it plainly did not—and the nearest near-miss is close enough to name.[1] Of the candidate designs we checked, every one fixes an axis, bundles both inside a method identity, or declines the decomposition. Our audited reference point is one such reported number made explicit: PGS reports per-task results in its Table 1 and the qualitative claim that surrogate class dominates, but computes no variance decomposition (Chemingui et al. 2024); we reproduce its Table 1 and compute the decomposition—$\eta^2_{\text{surr}} \approx 0.37$ on the uncorrected engine—then audit it. PGS could not itself have produced that decomposition or the interaction, because it fixes the surrogate as the operative factor and never crosses it: the crossed factorial is new territory, not a re-run of prior work.

The contribution. The instrument, and the confound set that running it exposed. The one-way analogue over model class is owned by fANOVA, computed at a single fixed search run (Hutter, Hoos, and Leyton-Brown 2014); the most thorough controlled surrogate-class comparison in Bayesian optimization varies seven classes but holds one acquisition rule fixed across every problem, runs online, and decomposes nothing (Li, Rudner, and Wilson 2024); the closest crossed grid reports descriptive metrics in online BO (Liang et al. 2021); and the closest two-axis design names fANOVA and *declines* it, on the grounds that one-factor-at-a-time analysis cannot detect interactions (Moosbauer et al. 2022). But an instrument inherits every confound in the grid it measures, and building the crossed grid surfaced five specific to offline-MBO surrogate comparison, each traceable to a line of the scoring path. The contribution is the composition: the five confounds, the protocol that removes them, and what the decomposition says once they are gone.

Contributions.

1. **Five named confounds and a removal protocol** (Sec-

[1]Table 3 of Tan et al. (2025) is a genuine 9×2 crossed grid in offline MBO, and four of its nine methods are clean optimizers applied to a trained model, so a 4×2 crossed sub-grid exists. It crosses training *loss* against method, never model class, and reports per-cell score and %gain rather than a decomposition. Decomposing its published table gives a loss axis far below its method axis (η^2 0.027 against 0.577); we state that only as an observation about that grid, whose method axis bundles model family with search paradigm and so is not commensurable with our optimizer axis. One title in our sweep names both axes—robust parameter design across three surrogate classes and two optimizer families (Elsayed and Lacor 2014)—and we could not obtain its primary text. If its design is a crossed factorial with a variance decomposition it is prior art; robust-parameter-design methodology is characteristically sequential, surrogates compared first and one winner then optimized, which would fail the crossing requirement. We state the conditional rather than omit it.

tions 3–4), each with a code-level trace, a signed effect on η^2_{surr}, and a pre-registered kill criterion; the β–σ criterion fired outright while the K-reversal criterion did not. This is the paper's novel object—no fANOVA-family study produces these confounds—and the crossed decomposition is the instrument that exposed them, not the contribution itself. Correcting the two *protocol* confounds moves η^2_{surr} *up* rather than down, from our decomposition of PGS's Table 1 ($\eta^2_{\text{surr}} \approx 0.37$) (Chemingui et al. 2024) to 0.405 $[0.290, 0.556]$—a direction we found no precedent for in the ML audit literature—while the corner intervals remain unresolvable at $n{=}7$ tasks; we claim the direction, not a level difference.

2. **A surrogate$\times$optimizer interaction, the term the design exists to estimate** (Section 4). A one-factor-at-a-time design cannot produce it: η^2_{inter} of 0.146–0.165, a bootstrap interval excluding zero in all four corners, four to thirty-three times the optimizer main effect. Its conditional effect and its marginal ($\eta^2_{\text{opt}}{=}0.038$) are different quantities—which is why "optimizer choice explains little *marginal* variance" and "optimizer choice is arbitrary" are not the same claim.
3. **An operating-point thesis** (Section 4): any η^2 headline is a joint artifact of five coordinates—K, β, search budget, engine corner, and the per-task normalizer—so a headline stating fewer is not reproducible. Two independent runs at one fixed operating point agree to 0.0018; leave-one-task-out moves the headline across 0.373–0.485.
4. **A mechanism narrowed by elimination** (Section 5): seven controls—σ, per-member width, query budget, held-out accuracy, mean roughness, premise coverage, and distance-to-support with surrogate inflation, the last three by pre-registered manipulations that refute accounts of our own—and none survives.
5. **A Design-Bench null with its power stated** (Section 6): no detectable difference at $n{=}7$ tasks, reported as such and never as equivalence (Agarwal et al. 2021).

Scope. No field reversal: that the statistical model often matters more than the acquisition heuristic is the organizing doctrine of the standard Bayesian-optimization review (Shahriari et al. 2016) and is a decade old, and our optimizer-axis result falsifies the *local* premise of one recent paper (Chemingui et al. 2024) and nothing wider. We claim no mechanism discovery, no equivalence on Design-Bench, and no result about ensemble size; Section 7 states each in full.

2 Background and Related Work

LCB, pessimism, and coverage. GP-LCB $\mu - \beta\sigma$ has regret guarantees under GP assumptions (Srinivas et al. 2012); deep ensembles (Lakshminarayanan, Pritzel, and Blundell 2017) provide a cheap uncertainty proxy but no calibration guarantee. The pessimism principle behind conservative offline methods is that a valid lower bound prevents the optimizer exploiting model error (Jin, Yang, and Wang 2021); we make that premise measurable. Conformal methods give distribution-free coverage, including inside the Bayesian-optimization loop (Angelopoulos and Bates 2023; Stanton, Maddox, and Wilson 2023); we use coverage of the LCB premise only as a *measurement*, taken where the optimizer actually operates, and Section 5 shows it is separable from optimization outcome—a caution against treating it as the mechanism.

Offline MBO and its evaluation. The conservative-surrogate lineage runs from Conservative Objective Models (Trabucco et al. 2021), which add a CQL-style regularizer (Kumar et al. 2020), through generative surrogates (Brookes, Park, and Listgarten 2019; Krishnamoorthy, Mashkaria, and Grover 2023); all hold the search rule fixed while varying the model. A neighbouring line varies the training *objective* at fixed architecture rather than the model class—the ranking-loss surrogate (Tan et al. 2025) beside the conservative regularizer (Trabucco et al. 2021)—and holds gradient ascent fixed throughout its main comparison. The diagnosis that oracle-based design queries its surrogate exactly where the surrogate's training data does not reach, so that its outputs and its uncertainty estimates alike become unreliable, is due to Fannjiang and Listgarten (2020); Section 5 reports where that account is sufficient and where it is not. Optimizer-as-contribution work (Chemingui et al. 2024) argues the converse but proposes a method rather than decomposing; its motivating premise—that offline black-box optimization has focused on improving surrogate models while using fixed search strategies—is the one *local* claim our design falsifies, and the only one. Design-Bench (Trabucco et al. 2022) standardizes tasks and, by fixing a protocol, documents the confounded status quo we dissect. The second half of our method—naming unreported choices, giving the protocol that removes them, showing the ranking move—is the established reality-check genre, whose shape we claim no credit for and whose canonical instances each pair a critique with a reusable tool (Henderson et al. 2018; Agarwal et al. 2021; Balduzzi et al. 2018; Ferrari Dacrema, Cremonesi, and Jannach 2019; Musgrave, Belongie, and Lim 2020; Lucic et al. 2018); in offline optimization SOO-Bench (Zhu, Qian et al. 2025) stress-tests optimizer *stability* rather than attribution. In online BO the corresponding controlled study crosses seven surrogate classes against a single fixed Monte-Carlo expected-improvement rule and reports no variance decomposition (Li, Rudner, and Wilson 2024)—broader than ours on the surrogate axis and, by construction, silent on both the search axis and the attribution between the two. Our study targets the surrogate$\times$optimizer confound, which neither that design nor any of the two-axis precedents the introduction names (Hutter, Hoos, and Leyton-Brown 2014; Liang et al. 2021; Moosbauer et al. 2022) decomposes.

3 Experimental Grid and Confounds

The grid. The crossed grid (Figure 1) runs over 7 synthetic tasks (Branin-2D through Griewank-30D, dataset drawn once at seed 0) at 30 seeds, plus 7 Design-Bench tasks at 16 seeds. **Surrogates:** a $K{=}5$ ensemble of two-hidden-layer ReLU MLPs (width 96, MSE, 35 epochs, Adam lr 3×10^{-3}), with μ, σ the member mean and standard deviation; an *exact GP* (BoTorch `SingleTaskGP` at its default covariance module, fit by marginal likelihood on a score-biased subsample, $N_{\max}$=800); and a *sparse variational GP* (SVGP; 128 inducing points, ARD Matérn-5/2, 250 ELBO steps, $N_{\max}$=2000). **Optimizers:** 100 Adam steps on x (lr 0.05, box-clipped); 5 rounds of Gaussian hill-climbing; and CMA-ES. All three surrogates are differentiable, so all nine cells are realizable. Every cell maximizes the LCB $\mu(x) - \beta\sigma(x)$ at β=2 and is scored by the ground-truth oracle on its 128 returned designs at the 100th percentile. Per task we min–max normalize the nine cell means and take a two-way ANOVA; η^2_{surr} and η^2_{opt} are the fractions of normalized-score variance carried by the two main effects, with task-and-seed hierarchical bootstrap intervals (B=10,000).

Two details of that description are themselves engine state: the exact GP's default covariance is ARD RBF under the BoTorch pinned for the synthetic grid but ARD Matérn-5/2 under the older version stamped on the Design-Bench runs, and the GPs alone spend per-run marginal-likelihood budget, which a matched-tuning arm freezes to zero for every surrogate (the surrogate effect retains 75% under that control). Neither is recoverable from a method description that does not stamp its engine.

Against that grid we name five confounds. Each is a specific line of the scoring path, not a judgment call, and each carries a signed effect (Table 1, Figure 2).

Confound 1: target scaling. The ensemble regresses on raw targets spanning -1560 to $+34$ while both GP surrogates z-score theirs (`mbo.py:34,157-162,181-187` against `mbo.py:293,379-381`). Correcting this alone moves η^2_{surr} from 0.367 [0.254, 0.559] to 0.283 [0.186, 0.444]. This is a measurement confound, not a bug report: the prior analysis states a min–max normalization that exists only in the analysis path, never in training, and the gap between stated and actual protocol is the finding. A second control reaches nearly the same value from a different axis: freezing every surrogate's per-run marginal-likelihood budget (the matched-tuning arm above) retains 75% of η^2_{surr}, landing near the 0.283 target scaling reaches. We did not run the two jointly, so whether they are independent corrections or two views of one entanglement is something our data cannot settle.

Confound 2: candidate/oracle protocol. Two of three optimizers propose 256 designs and report the oracle-selected best 128, while the third proposes 128 (`mbo.py:35-40,249-255,269-271`). The reported percentile is therefore two different estimands in one column. Correcting this alone moves η^2_{surr} from 0.367 to 0.450 [0.312, 0.649]. This is the strongest of the five: the protocol the prior analysis declares it holds identical is the identifiability license for the whole comparison, and the license is false.

Confound 3: ensemble size K. K enters as a confound to be controlled, never as a mechanism, and has two halves that must be read together. It is *sensitive*: η^2_{surr} is 0.326, 0.366, 0.408, 0.389 at $K = 2, 3, 5, 10$, peaking at exactly the K=5 the field fixes by convention (Lakshminarayanan, Pritzel, and Blundell 2017) (Figure 2). And it does *not* reverse: at K=2 the ensemble's normalized cell marginal is 0.283 against the exact GP's 0.767 and the SVGP's 0.739, so our own pre-registered kill criterion—"if the ensemble's marginal at K=2 exceeds the GP's, the surrogate main effect is a K artifact and the headline reverses"—did not fire. We also report that the figure this criterion was written against, a task-normalized ensemble score of 0.95 at K=2 in the prior supplement, *does not reproduce* on the audited engine, which gives 0.283 at that cell. (These two 0.283s are unrelated quantities: Confound 1's is a variance-explained statistic, this one is a normalized cell marginal. Neither corroborates the other.) Our sweep runs over $K \in \{2, 3, 5, 10\}$ against the $K \in \{2, 5, 10\}$ over which ensemble surrogates were found robust in online BO under one fixed acquisition rule (Li, Rudner, and Wilson 2024), so we share three of its four points and add only K=3. We therefore claim no priority on the range. The residual is that the two sweeps measure different quantities: that work measures whether ensemble-BO *performance* is robust to K and reports that it is, while we measure whether the variance *attributed to the surrogate axis* is robust to K and report that it is not. Both can hold at once, and only the second bears on a decomposition—a further reason to treat K as a nuisance coordinate, not a result.

Confound 4: β–σ mismatch. We registered that a shared β=2 delivers different effective pessimism to different surrogate classes, with the kill criterion "if the ratio is $\sim$1, the acquisition was matched and this confound does not exist." *That kill fired.* The median $\sigma_{\text{GP}}/\sigma_{\text{ens}}$ is 1.19 at matched K=5 (1.21 pooled over K), so the fifth unmatched hyperparameter does not exist grid-wide, and we withdraw the aggregate claim. What survives is per-task: the ratio spans 0.07 on Branin-2D to 1.44 on Styblinski-5D and Rastrigin-15D, a 20-fold spread that a single shared multiplier cannot track. The general principle that a fixed multiplier on an uncalibrated interval is not comparable across model classes is owned by the interval-validity literature (Dewolf, De Baets, and Waegeman 2022); a sharper and different result sits beside it in offline RL, where a training-procedure choice *within* one surrogate class—sharing pessimistic targets across ensemble members rather than keeping them independent—can *invalidate* nominal pessimism outright rather than merely miscalibrating it, flipping the lower bound's sign so that a nominally pessimistic ensemble becomes optimistic (Ghasemipour, Gu, and Nachum 2022). That is a within-class result, not a cross-model-class instance, and we do not read it as one. We claim only the offline-MBO acquisition-comparison application, in its narrowed per-task form.

Confound 5: search-intensity budget. Surrogate-query budgets are unmatched by 11.8× across the optimizer axis. The counts are *measured* by a counting proxy rather than derived: gradient 51,456, perturbation 4,352, CMA-ES median 6,528 (range 932–6,536). Three corrections belong here. The previously estimated "6×–59×" spread was never measured, and CMA's nominal budget of 3000 is a cap that rarely binds because the library stops on its own tolerances first. The proposal-count half of this confound is already removed by Confound 2, so the surviving inequality is in surrogate queries alone and does not double-count. And the trust region flagged in our own audit is inactive in the main grid, so equalizing queries isolates budget. Matching every optimizer

Table 1: The five confounds. Signed effect is on η^2_{surr} (synthetic, n=7, 30 seeds) except Confound 5, which acts on η^2_{opt}. The two *protocol* confounds move the headline against the ensemble-size confound; the net is Section 4.

Confound	What it breaks	Signed effect
1. Target scaling	ensemble trains on raw y	$0.367 \to 0.283$
2. Candidate/oracle	two estimands in one column	$0.367 \to 0.450$
3. Ensemble size K	headline fixed at the peak	0.326–0.408
4. β–σ match	per-task, not aggregate	ratio 0.07–1.44
5. Query budget	optimizer = search effort	$0.005 \to 0.038$

at Q=51,456 (achieved 51,200 / 51,456 / 51,472; 0 of 630 cells more than 5% off target—the budget-matched synthetic arm runs 10 seeds per cell, not the 30 of the main grid, so $630 = 7 \times 9 \times 10$) moves the optimizer main effect from our own unmatched η^2_{opt} of 0.005 to 0.038 $[0.003, 0.123]$—an eightfold understatement of our own baseline, which we disclose. (The prior analysis reported 0.01 at this cell; our unmatched reproduction gives 0.005.) The same matching was run on Design-Bench, where it *raises* the optimizer effect rather than removing it (Section 6), so this confound is now measured out of both suites. What the magnitudes license is deferred to Section 7.

4 De-Confounding Protocol and Results

The protocol. Standardize the ensemble's regression targets. Equalize the candidate proposal count across optimizers and impose one selection rule, so the oracle never chooses the reported set. Hold K, β and the query budget fixed and *report them*. Stamp the engine—library versions, flags, commit—with every number. Switching the first two on and off gives four corners (Table 2).

The correction strengthens the surrogate effect. The named audits in this genre each report a claimed advantage shrinking or reversing: an LSTM baseline overtaking newer architectures once hyperparameters are controlled (Melis, Dyer, and Blunsom 2018), the GAN comparison (Lucic et al. 2018), the metric-learning reality check (Musgrave, Belongie, and Lim 2020) and the neural-recommender audit (Ferrari Dacrema, Cremonesi, and Jannach 2019). That is an observation over instances, not a law about the genre, and we use it only as such. This audit strengthens the effect instead. Correcting the two measurement and protocol confounds moves the rewarded surrogate advantage *up*—from our decomposition of PGS's Table 1, $\eta^2_{\text{surr}} \approx 0.37$ (Chemingui et al. 2024), to a corrected 0.405 $[0.290, 0.556]$; the ensemble-size confound moves it the other way, so the strengthening is the net of the two protocol confounds and not a property of de-confounding in general. We searched the ML reality-check and reproducibility literature for a de-confounding audit reporting a corrected variance-explained statistic above its own published value and found none. Two partials must be pre-empted. An ImageNet replication reports a relative slope above one—1.69 on CIFAR-10, 1.11 on ImageNet—but that is a relative slope, not a variance-explained scalar audited against its own prior value, and its

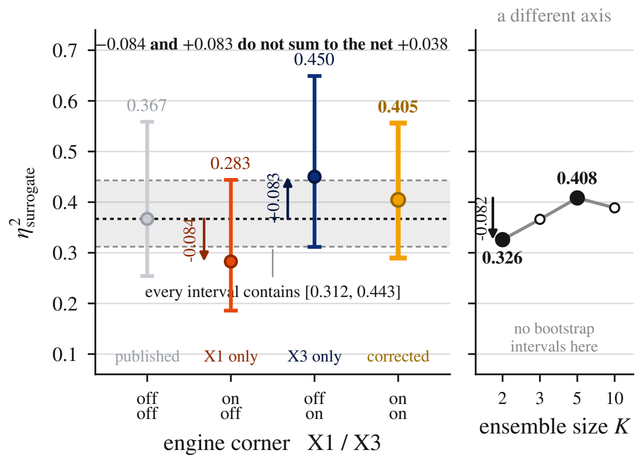

Figure 2: The four engine corners as measured levels, not as a composing cascade. Target scaling alone (X1) moves η^2_{surr} *down* 0.084 and the candidate/oracle rule alone (X3) moves it *up* 0.083, but the corner where both are applied sits 0.038 *above* the uncorrected one: the corrections are *not additive*, and the corrected 0.405 $[0.290, 0.556]$ is a measured joint level rather than a sum of the separate effects, which very nearly cancel. The shaded band is the region every corner's bootstrap interval contains—the four intervals overlap so heavily that no corner is separable from another at n=7, which is the pre-registered KB5 result. What this figure localizes is therefore the *signed direction* of each correction, never an additive budget and never a corner-to-corner difference. Ensemble size sits in a separate panel because it is a confound on a different axis, not a fifth corner; its sweep was not bootstrapped, so those points carry no intervals. Confounds 4 and 5 are absent by measurement, not omission: Confound 4's aggregate kill criterion fired, so it has no grid-wide η^2_{surr} effect, and Confound 5 acts on η^2_{opt}. Synthetic grid, n=7 tasks, 30 seeds, B=10,000 bootstrap.

own headline is an 8–11 point accuracy drop, squarely in the shrink family (Recht et al. 2019); and stratified reanalysis of deep-RL results moves effects both ways (Agarwal et al. 2021). Nor is the direction statistically surprising: a removed confound can as easily have been suppressing an effect as manufacturing one (Bressan 2019), and where the question has been studied at scale—500 primary effect sizes recomputed across 33 meta-analyses—corrections run up 19 times against down 14, with no systematic bias in either direction (Maassen et al. 2020). An upward correction is therefore unusual in this specific ML genre and unremarkable as a statistic. The claim is that it is unprecedented as an instance, not anomalous.

Corner resolution at n=7 tasks. We pre-registered that it would not be, and the prediction confirmed. The four corner intervals have widths 0.26–0.34 against a corner range of only 0.167, and they share a common region, $[0.312, 0.443]$, that every one of them contains (Figure 2). So 0.405 never appears in this paper without its interval, and no corner is asserted to differ from another. The bootstrap is validated against the reproduction target's precision: our uncorrected corner's interval width, 0.305, matches the 0.32 recorded for it (Chemingui et al. 2024).

The β-dependence is the firm result. η^2_{surr} rises monotonically with pessimism—0.184 $[0.085, 0.354]$, 0.214, 0.282, 0.406 $[0.285, 0.564]$, 0.408 at $\beta = 0, 0.5, 1, 2, 5$—and the β=0 and β=2 intervals barely overlap, where the four corner intervals do not separate at all (Figure 3). This is the axis on which the seven tasks come closest to separating a difference the corner decomposition cannot resolve at all, and we lead the decomposition with it for that reason. The query budget goes further still: η^2_{surr} more than doubles from 0.243 at Q=4,352 to 0.526 at Q=51,456, and those two intervals are *disjoint*—$[0.189, 0.355]$ against $[0.421, 0.719]$—the only outright separation anywhere in this decomposition, and it comes with a ranking flip, gradient leading at the low budget and perturbation at the high one. That axis is not ours to claim as new. Budget-dependent benchmarking is a named methodological category with a decade-old apparatus, which prescribes fixed-budget and fixed-target views precisely because a single assessment cannot predict performance at a different budget (Hansen et al. 2021), and the qualitative statement that bad models can outperform good models given enough budget was made at NeurIPS in 2018 (Lucic et al. 2018). What we add is the measured instance on a new axis: the surrogate-versus-optimizer version, quantified, with disjoint bootstrap intervals and a ranking flip where the prior statements are qualitative. The *magnitude* of any headline η^2_{surr} is therefore a joint artifact of five chosen operating-point coordinates—K, β, the query budget, the engine corner, and the per-task normalizer—so a headline that does not state all five is not reproducible. We add the fifth explicitly because every η^2 here is computed on per-task min–max normalized cell means whose range is set by two of the nine arms under comparison, an endogenous form that is known to be outlier-exploitable (Jordan et al. 2024), documented flipping a ranking at its origin (Bellemare et al. 2013), and pool-dependent in the way rank statistics also are (Balduzzi et al. 2018). Section 7 reports what changes when it is varied. This strengthens the decomposition rather than weakening it, because *within this synthetic grid* the *direction* is invariant to all four: across every β, K and corner we ran, the GP marginals sit at 0.66–0.85 and the ensemble's at 0.17–0.35, never crossing—the smallest gap between the two is 0.32. Only the size of the gap moves. That invariance is a statement about these seven synthetic tasks and does not travel across suites: on Design-Bench the same direction is not detectable at this power (Section 6).

The interaction is the term the design exists to estimate. The fourth column of Table 2 is the one a one-factor-at-a-time study cannot produce, and we report it rather than print it. η^2_{inter} is 0.165, 0.146, 0.152 and 0.160 across the four corners, with bootstrap intervals $[0.107, 0.260]$, $[0.088, 0.249]$, $[0.094, 0.230]$ and $[0.087, 0.236]$: the interval excludes zero in all four, at four to thirty-three times the optimizer main effect. It survives the same bias correction the main effects need, at 0.134–0.156. And it is the most stable quantity in this paper—its range across corners is 0.018 against the headline's 0.167, so the coordinate we show the headline is most sensitive to barely moves it.

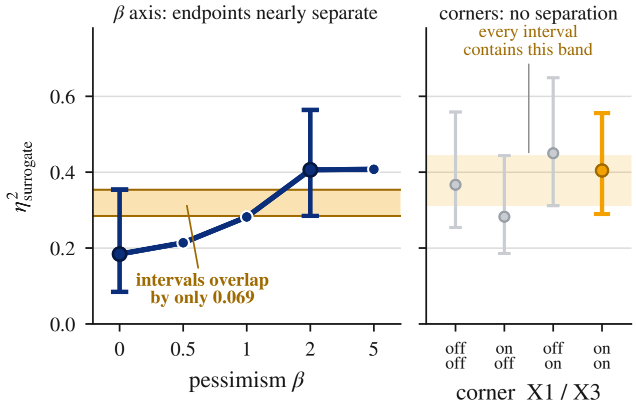

Figure 3: The β axis is the one these seven tasks come closest to resolving, and the engine corners are the one they cannot resolve at all. Left: η^2_{surr} rises monotonically with pessimism, and the β=0 and β=2 intervals overlap by only 0.069. Intervals were computed at those two endpoints and are drawn nowhere else. Right: on the same scale, the four corner intervals overlap completely, sharing a common region every one of them contains. This is why the paper leads its decomposition with β and never asserts a corner difference. Synthetic grid, 7 tasks, 30 seeds, B=10,000.

Table 2: Four-corner decomposition (synthetic, 7 tasks, 30 seeds, B=10,000 bootstrap). X1 standardizes the ensemble's targets; X3 equalizes the candidate/oracle protocol. The intervals overlap heavily: corner differences are *not* resolvable at n=7.

X1 / X3	η^2_{surr} [95% CI]	η^2_{opt}	η^2_{inter}	Friedman p
off / off	0.367 $[0.254, 0.559]$	0.013	0.165	6.1×10^{-5}
on / off	0.283 $[0.186, 0.444]$	0.036	0.146	1.7×10^{-3}
off / on	0.450 $[0.312, 0.649]$	0.006	0.152	4.1×10^{-5}
on / on	**0.405** $[0.290, 0.556]$	0.005	0.160	8.1×10^{-4}

Three things follow, and a fourth bounds them. First, this is what the crossed design buys: one-factor-at-a-time experimentation cannot estimate interactions at all, which is the standard argument for a factorial (NIST/SEMATECH 2012; Box, Hunter, and Hunter 1978), and it is why the closest two-axis precedent names the analysis and declines it (Moosbauer et al. 2022). Second, it is the missing evidence for a claim we make elsewhere: a small η^2_{opt} licenses "optimizer choice explains little *marginal* variance" and never "optimizer choice is arbitrary," and the interaction is why—optimizer choice has a small marginal effect and a substantial conditional one, by four to thirty-three times. That pattern is textbook rather than anomalous: a factor can be the least important main effect while appearing in almost every significant interaction, and large interactions can obscure main effects (NIST/SEMATECH 2012). Third, we concede the precedent in its defensible form. Interaction reporting is routine in-

Table 3: Normalizer robustness. Four engine corners $\times$ three per-task normalizers, recomputed from the stored per-seed grids. $\eta^2_{\text{surr}} > \eta^2_{\text{opt}}$ in all twelve; $\eta^2_{\text{inter}} > \eta^2_{\text{opt}}$ in eleven, inverting only in the marked row.

Corner	Norm.	η^2_{surr}	η^2_{opt}	η^2_{inter}	int/opt
off / off	min–max	0.367	0.013	0.165	12.4×
off / off	z-score	0.426	0.019	0.193	10.0×
off / off	rank	0.487	0.050	0.053	1.1×
on / off	min–max	0.283	0.036	0.146	4.0×
on / off	z-score	0.347	0.042	0.189	4.5×
on / off	rank	0.318	0.075	0.049	**0.6**×
off / on	min–max	0.450	0.006	0.152	26.9×
off / on	z-score	0.494	0.008	0.167	20.6×
off / on	rank	0.524	0.021	0.062	2.9×
on / on	min–max	0.405	0.005	0.160	32.8×
on / on	z-score	0.454	0.003	0.179	71.3×
on / on	rank	0.388	0.017	0.071	4.1×

side the functional-ANOVA tradition this design descends from, which decomposes into main effects and pairwise interactions by construction (Hutter, Hoos, and Leyton-Brown 2014); what we find no instance of is an interaction reported and interpreted between *surrogate model class* and *search routine* as two independently manipulated pipeline components.

The bound is that this is an existence-and-ordering claim under a disclosed normalizer, not the stable quantity "0.15." Recomputing the full decomposition across four corners and three normalizers (Table 3), $\eta^2_{\text{surr}} > \eta^2_{\text{opt}}$ holds in *all twelve* combinations and η^2_{surr} is largest in all twelve, so the headline ordering is untouched; the interaction exceeds the optimizer main effect in *eleven* of twelve, by $1.1\times$ to $71.3\times$, and inverts in one—the on/off corner under rank normalization, which is the corner carrying the largest optimizer main effect under every normalizer. Rank transformation roughly halves the interaction throughout. The second bound is suite-specific: on Design-Bench η^2_{inter} is only 0.007–0.036 and subordinate to η^2_{opt} there, and we do not have an explanation for the failure to generalize. Frozen cells are a candidate—a cell returning a bit-identical constant can carry neither a main effect nor an interaction—but Design-Bench's own η^2_{opt} of 0.05–0.17 shows the axes are not all flattened, so we state the scoping as an unexplained limit rather than as a resolved one.

5 Isolating the Source of the Gap

We ran seven controls, each capable of explaining the GP–ensemble gap, and none does. Two were built to confirm accounts of our own and refuted them instead. Each control's scope travels with it below, because a bare count of seven would claim more than two of them establish: capacity is eliminated at fixed depth, and σ is eliminated at this ensemble construction.

On our tasks, σ tracks distance more than error. The ensemble's σ correlates with pointwise absolute error at only $\rho \approx 0.07$, but with k-NN distance to the training data at $\rho \approx 0.26$—three to four times larger, positive on all seven tasks, and above 0.28 on the four tasks between 5 and 15 dimensions (it falls back to 0.20 and 0.23 at 20 and 30). The prior analysis concluded σ was uninformative by measuring it against the wrong target; this is the corrected measurement. We state it as a scoped observation rather than a dichotomy: distance-awareness and error-tracking are not mutually exclusive, and in-distribution regression work reports ensemble uncertainty correlating with error an order of magnitude more strongly than we measure here. Read against the distance-aware literature the result is a *contrast* rather than a corroboration. Deep ensembles are precisely the model class that work characterizes as *not* distance aware, assigning low uncertainty to points far from the data (Liu et al. 2020; van Amersfoort et al. 2020); ours shows a modest but positive distance correlation instead. Two scope gaps keep that honest: those results validate on classification with no regression experiments, against a study that is regression throughout, and they measure geometry we do not reproduce.

One alternative explanation for $\rho \approx 0.07$ we did not examine and do not exclude. Our σ is the empirical standard deviation across K members trained to minimize squared error, which is the construction contrasted against a likelihood-trained heteroscedastic ensemble and reported to underestimate predictive uncertainty—an 80% nominal interval covering roughly 20% of observations (Lakshminarayanan, Pritzel, and Blundell 2017). The low error correlation may therefore be a property of this construction rather than of ensembles generally. We did not run the likelihood-trained variant, so Elimination 1 below removes σ *as constructed here*, and a different construction is an open arm rather than one we have closed. We do not claim that squared-error ensembles are broken.

Elimination 1: it is not σ. With pessimism removed entirely (β=0, pure posterior-mean maximization), 61% of the GP's advantage survives and the interval excludes zero: the marginal gap is 0.319 $[0.196, 0.460]$ at β=0 against 0.525 $[0.406, 0.614]$ at β=2. Both conditions are read on one β-invariant normalizer, a per-task min–max fit once over the pooled cell means of both and refit inside each bootstrap resample. The pessimism increment is itself significant—0.203 $[0.007, 0.396]$, p=0.020—so the statement is base-plus-amplification: pessimism *amplifies* a mean-quality base rather than creating it—though it passes narrowly, the surviving base of 0.319 clearing its pre-registered collapse threshold of 0.263 by a margin inside the bootstrap interval—and any claim that the advantage is independent of pessimism is refuted by our own data. That increment is a *paired bootstrap delta*, resampled jointly, not a difference of the two marginals: subtracting the point estimates gives 0.206, and the 0.203 is the correct paired quantity rather than an arithmetic slip. The cross-check is the strongest single evidence we have that these numbers hold. Recomputed on an entirely different normalization—a per-task z-score by the task's own pooled cell-mean standard deviation—the same two conditions give 0.904 against 1.473 standard deviations. The ratio is 0.614 there against 0.608 on the min–max scale: an independent recomputation on a different scale landing within 0.006 of the same ratio, which is what makes this a measurement rather than a normalization artifact.

Elimination 2: it is not finite width, at fixed depth. The gap could be an under-parameterization artifact: wide networks approach GP behaviour, at initialization and under gradient descent alike (Jacot, Gabriel, and Hongler 2018; Lee et al. 2018, 2019), and finite networks show spectral bias (Rahaman et al. 2019). Our K-sweep does *not* test that—K is cardinality, not per-member capacity, and offering it here would be a category error—so we ran the ablation that does. The nearest existing architecture ablation runs the other way, *shrinking* the network and finding the surrogate ranking mostly unchanged (Li, Rudner, and Wilson 2024); the width objection predicts the gap closes as the net *grows*, so we sweep upward, at fixed K, varying nothing else. At fixed $K{=}5$, sweeping per-member width over a $10.7\times$ range leaves the gap unchanged (Figure 4, top): 0.480 $[0.365, 0.576]$ at $w{=}96$ against 0.476 $[0.208, 0.647]$ at $w{=}1024$, 99.1% retained, shrinkage -0.006 $[-0.210, 0.161]$. The curve is flat with noise and non-monotone—it dips to 0.336 at $w{=}256$ and returns—not a decay. The shrinkage is a paired bootstrap quantity, so its -0.006 need not equal the 0.004 difference of the two endpoint point estimates. Three limits are ours. The interval widens monotonically with w ($0.211 \to 0.439$), making $w{=}1024$ the least precise point on the curve, so the supported claim is that the gap *does not close*, not that it is identical there. The sweep stops at 1024 while the kernel limits are asymptotic, so this answers the objection at practical widths and makes no asymptotic claim. And the third is a scope limit on the motivation rather than on the measurement: spectral bias is reported as primarily a *depth* phenomenon, its own controlled ablation finding that increasing depth at fixed width substantially improves the fitting of higher frequencies while increasing width at fixed depth helps considerably less (Rahaman et al. 2019). Our ensemble is a two-hidden-layer MLP throughout this sweep, so we vary the lever that source calls weak and hold fixed the one it calls dominant. The width-driven form of the capacity objection is answered; the *depth*-driven form remains open, and a depth sweep at fixed width is the experiment that would close it. As a validity check, $w{=}96$ reproduces the incumbent grid bit-exactly for the gradient and CMA cells at seed 0 on four of the seven tasks (Levy, Rosenbrock, Ackley, and Griewank).

Elimination 3: it is not predictive accuracy. The ensemble is the *more* accurate surrogate. Its held-out normalized RMSE on a 20% split never seen by the grid is below the GP's at every width (Figure 4, bottom)—$0.445, 0.405, 0.393, 0.388$ at $w = 96, 256, 512, 1024$ against the GP's 0.479—winning in 26 of 28 (task, width) cells and on all seven tasks at $w \geq 512$. And it still loses the optimization comparison by roughly 0.48. The registered form was narrower, and we report it as primary: read only at the two cells where ensemble and GP RMSE are statistically indistinguishable (Styblinski-5D at $w{=}256$, difference CI $[-0.021, 0.028]$, and at $w{=}512$, $[-0.043, 0.010]$), the mean gap is still 0.375. The $7/7$ result is a post-hoc strengthening, labelled as one. Held-out NLL is not usable as a second accuracy axis—the GP's mean is some $32\times$ the ensemble's, but it is carried almost entirely by one task and reverses on three others—because that is a

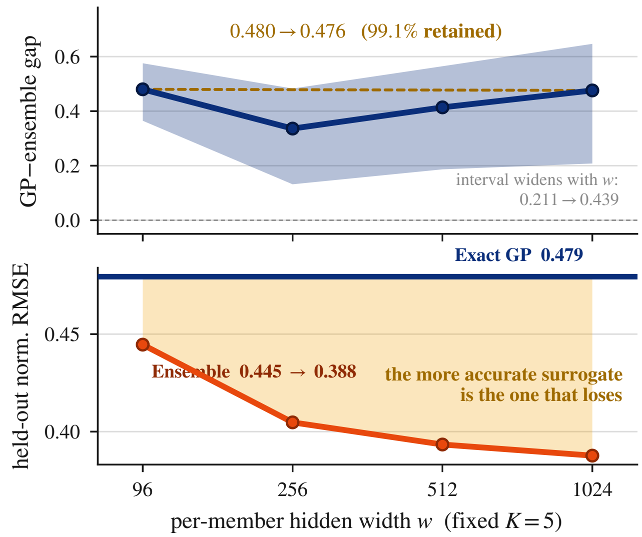

Figure 4: The two eliminations that rule out capacity and accuracy, in one image. Top: at fixed $K{=}5$ the GP–ensemble gap does not close over a $10.7\times$ width sweep ($0.480 \to 0.476$); the curve is non-monotone, and the bootstrap band widens with w, so $w{=}1024$ is the least precise point on it—a limit we state rather than smooth over. Bottom: the ensemble's held-out normalized RMSE falls monotonically with width and lies below the exact GP's flat line at every width. The more accurate surrogate is the one that loses. Synthetic grid, 7 tasks, 30 seeds, $B{=}10{,}000$; RMSE on a 20% split the grid never saw.

calibration finding, orthogonal to this one. *The more accurate surrogate is the one that loses*, which removes predictive error from the candidates.

Elimination 4: it is not budget, and more budget makes it worse. Equalizing surrogate queries does not close the gap; it widens it, and the composition of that widening matters: the ensemble's marginal *falls* as budget rises ($0.361 \to 0.240$) while both GPs rise ($0.755 \to 0.849$ and $0.713 \to 0.794$). More search pressure finds more of the off-distribution maxima the ensemble's mean admits. This was not what the budget arm was designed to test, so it is an observation, not a pre-registered result, and corroborates a mechanism rather than any η^2 magnitude.

Eliminations 5 and 6: it is not how rough the mean is, and it is not premise coverage. Our previous account named the roughness of the surrogate's mean as the axis. We tested it bidirectionally under pre-registration and it failed. Penalizing the ensemble's mean gradient cuts its measured roughness by up to 98.0% (spectral normalization, 90.3%). The gap did not close; it *widened* for every variant (base 0.306 against 0.322–0.552), no closing interval excludes zero, and the Lipschitz-bounded ensemble is markedly worse in score ($0.657 \to 0.537$ at $w{=}96$, $\to 0.346$ at $w{=}1024$). A self-test planting synthetic ground truths confirms the analyzer returns the opposite verdict when smoothing genuinely closes a gap. Smoothing fixes the *premise* and not the outcome—own-proposal coverage rises monotonically to 0.998 under the strongest penalty with no effect on the gap—so the coverage proxy is separable from what it was meant to explain, corroborating on a manipulated axis the modest $\rho(\text{coverage}, \text{score}) = 0.29$ we measure across cells. The complementary arm, roughening the GP, is void: no kernel setting raised the fitted mean's roughness by the registered 25% (best $+12.6\%$), because a posterior mean conditioned on $\sim$800 points is smooth from the conditioning rather than the kernel—roughness and fit quality are not independently manipulable in a fitted GP.

Elimination 7: it is not distance to the data, and it is not surrogate inflation. The next hypothesis is that the ensemble's optima sit further off-support, where its mean is free to invent maxima, and that it over-predicts them more. We pre-registered both limbs and instrumented every returned optimum with its 10-NN distance to the offline data and its over-prediction against the oracle (168 cells, 30 seeds, reproducing the incumbent grid bit-exactly at seed 0). Both come back null: the ensemble's optima are $+0.116$ $[-0.004, 0.333]$ further out and $+0.043$ $[-0.652, 1.085]$ more inflated, both intervals covering zero, and the distance estimate is one task: Branin-2D contributes $+0.743$ while the other six lie between -0.022 and $+0.060$. The loss is not null: at those same optima the ensemble is worse in *true oracle value* by -1.406 $[-2.803, -0.375]$ in units of $\mathrm{sd}(y_{\mathcal{D}})$, excluding zero, and binned on distance it is worse in four of five bins and tied in the fifth. The landscape law behind those bins is real—across 5,040 optima, distance predicts both over-prediction (ρ=+0.758) and true loss (ρ=−0.818)—but it does not discriminate (Figure 5): the classes sit at the same median distance (0.87 against 0.86) and differ in outcome anyway. Off-support maxima are inflated and worthless; that is a property of the problem, not of the ensemble.

That law is not ours, and stating whose it is sharpens what we add. Fannjiang and Listgarten (2020) articulated it in 2020: oracle-based design queries the oracle in regions the oracle's training data does not represent, so that its outputs, *including its uncertainty estimates*, become unreliable beyond the training data. Our measurement confirms that account in aggregate and shows it is *insufficient* for the question this paper asks. Off-support degradation is a property of the problem shared by every surrogate class here; it therefore cannot explain a difference *between* classes, and at matched median distance (0.87 / 0.86 / 0.84) the ensemble is still worse in true oracle value by -1.406 $[-2.803, -0.375]$. The refinement is that the off-support story is right about where surrogates fail and silent about which maximizers each class admits once it gets there. The same distance-graded degradation has a quantitative form in a different setting, where over-optimizing a learned proxy degrades true reward along closed-form curves whose coefficients depend on the proxy's class (Gao, Schulman, and Hilton 2023); our Elimination 7 is consistent with that shape and moves it onto a new axis—architecture class rather than model scale—rather than contradicting it.

The second limb is uninformative rather than negative, and the distinction is ours to make. It proposed raising the GP's prior mean until its far field stopped reverting to the data mean. The manipulation does not move the quantity it targets: a constant above every observation shifts the far-field posterior mean by under 1% of the distance to it (R=0.009 against a registered 0.25), because the fitted lengthscales are of order the domain diameter and no point in the feasible cube is in a reversion regime. The one arm that does remove reversion carries $211\times$ the incumbent's held-out RMSE, and a surrogate that has stopped predicting is not evidence about reversion. A premise failed rather than a prediction surviving—which establishes that here any account of the GP's advantage resting on prior reversion is unavailable, not merely untested.

The gap survives seven controls. Seven candidate explanations now have a control behind them and none survives: σ, per-member width, query budget, held-out accuracy, mean roughness, premise coverage, and distance-to-support with surrogate inflation. The seventh adds a positive localization: the loss is real and sits at the returned optimum's oracle quality, while every landscape property we can construct to explain it is matched between the classes. What differs is *which* off-distribution maximizers a surrogate's mean admits: not how rough those means are, not how far out the optima lie, and not how much they over-predict. That is a diagnosis, not a mechanism with a positive causal test behind it, and we label it as such. A skeptic could read this localization and Section 4's interaction as one phenomenon counted twice—the ensemble's bad basins are reachable by gradient ascent and missed by perturbation, which *is* the interaction. The defence is that these surrogate-geometry measurements do not read off the optimizer axis: the σ-against-distance correlation is a property of the surrogate alone, and the matched-distance oracle-value gap is pooled across all optimizers rather than traced to any one, whereas the interaction is a surrogate$\times$optimizer decomposition. The mechanism is therefore one phenomenon triangulated by two independent instruments—an optimizer-agnostic geometry measurement and a factorial decomposition—not the interaction restated.

A question the grid poses and we test: is the advantage conditional on the optimizer? The interaction says the surrogate gap depends on which optimizer is paired with it. In raw oracle units—no normalization and no bootstrap, so neither the normalizer critique nor the small-n η^2 bias reaches this—the gap between the exact GP and the ensemble is smaller under perturbation than under *both* gradient ascent and CMA on all seven tasks, spanning 2D to 30D, averaging under a tenth of the larger aggressive-optimizer gap. On the audited engine those per-task perturbation gaps run 0.053, 5.996, 0.047, 0.008, 0.187, -0.004 and 25.556 against gradient's 6.95, 15.13, 0.46, 0.09, 4.92, 5.10 and 288.46; on Ackley-20D the gap has closed entirely rather than merely shrunk.

We report that as a question rather than a result, because the obvious competing explanation is not excluded and our one discriminating test points toward it. A floor effect—perturbation simply underperforming and compressing all three surrogate cells toward a common achievable value—would produce this same shape. The case that discriminates is Styblinski-5D, the one task where perturbation is the grid's

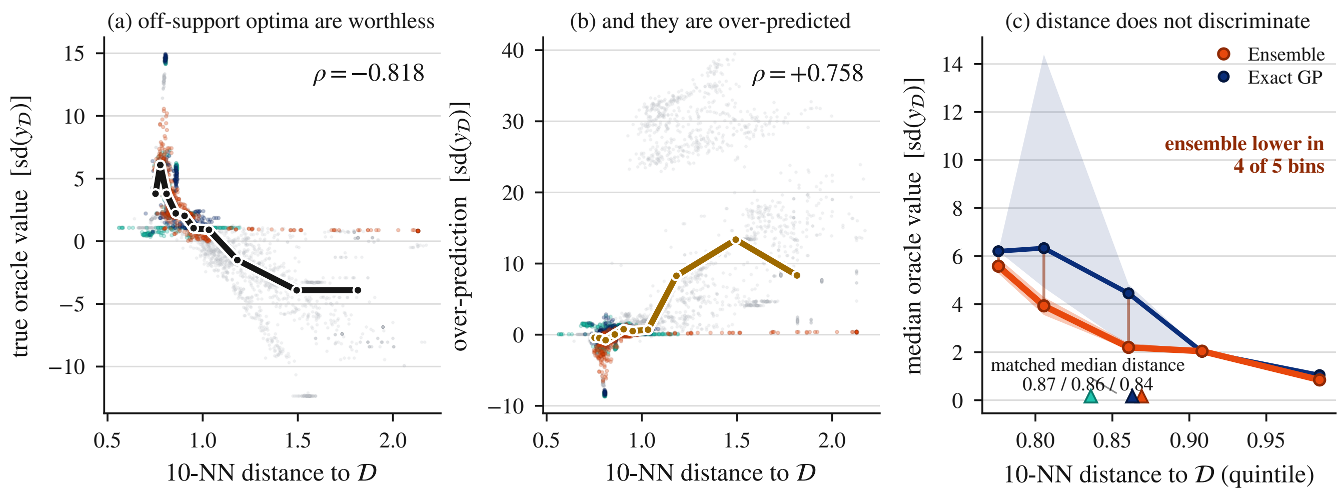

Figure 5: The seventh elimination: the loss is real, it sits at the returned optimum's oracle quality, and distance does not explain it. Every returned optimum is instrumented with its 10-NN distance to the offline data ($5{,}040$ optima $= 7$ tasks $\times$ 24 arms $\times$ 30 seeds; the five gray arms are the prior-mean probes, which enter the pooled statistic but carry no surrogate-class meaning). (a) Distance predicts worthlessness, $\rho{=}-0.818$ pooled, and (b) it predicts over-prediction, $\rho{=}+0.758$: off-support maxima are inflated and worthless. (c) But that law does not discriminate between classes—they sit at matched median distance (0.87 / 0.86 / 0.84) and the ensemble is still lower in oracle value at matched distance, in four of five bins. Panel (c) is the ensemble-versus-exact-GP comparison the elimination states; SVGP is omitted from the bin trace because its own distance–value correlation is near zero ($\rho{=}-0.09$) where the ensemble's is -0.83 and the exact GP's -0.76. Bands are 95% bootstrap intervals on the bin median. Audited engine, X1 and X3 on, $\beta{=}2$, $K{=}5$.

best optimizer (GP with perturbation at 35.9), and it carries by far the *weakest* attenuation, at 40% of the aggressive gap against at most 8.8% everywhere else. That is what a floor effect predicts and not what a genuine neutralization predicts. One contrary point at $n{=}7$ does not settle the matter, but it is the only case that speaks to it and it speaks against the causal reading. The descriptive fact is solid; the mechanism is not. On this synthetic grid, ensembles paired with gradient ascent produced systematic inversions (Figure 6) while pairing them with perturbation attenuated the gap; whether that reflects a genuine surrogate$\times$optimizer interaction or the floor effect above is unresolved.

The acquisition ranks its own hallucinations above designs it holds. What "admits an off-distribution maximizer" costs in practice is measurable directly, because the candidate/oracle correction puts the starting designs *inside* the returned pool: every cell is seeded with the top-128 designs of the offline data plus lightly perturbed copies, and gradient and perturbation optimizers pool every iterate together with that seed set before returning the top 128 by surrogate LCB. A cell that returns something worse than what it was handed has had its own acquisition rank a design it invented above a real design it was already holding. We call that an inversion and count it per seed (Figure 6). It is close to ensemble-specific: on the audited engine the ensemble inverts on at least one optimizer on all seven tasks, against three of seven for the SVGP and two of seven for the exact GP, and where the GPs invert at all they do so on tasks the ensemble inverts on more heavily. One of the sharpest cells is Branin-2D under ensemble gradient ascent—six other ensemble and SVGP cells tie it, seven in all meeting the criterion of unit inversion rate and unit mean fraction worse—where all 30 seeds invert and, on average, *every one* of the 128 returned designs scores below the best design the cell started with. This is the elimination-7 localization made concrete—the loss sits at the returned optimum's oracle quality—and it is a demonstration within this grid, descriptive at $n{=}7$ tasks, not a mechanism with a causal test behind it.

What it is an instance of has a name. Offline RL formalizes the requirement that a policy learned from a fixed dataset should not be worse than the baseline it was given, with high probability, as *safe policy improvement*, and measures algorithms by how often they violate it (Thomas, Theocharous, and Ghavamzadeh 2015; Laroche, Trichelair, and Tachet des Combes 2019); the conservative-value lineage supplies the complementary shape, in which the learned value function widens rather than inverts the gap between in-distribution and off-support designs (Kumar et al. 2020). Offline MBO optimizers ship with no analogous guarantee, and the inversion is what that absence looks like when measured: an acquisition returning designs worse than the ones it was handed is a safe-improvement failure, not merely an awkward number.

Gradient collapse on the audited engine. Our own released tuning sweep concluded that the ensemble's gradient collapse is an artifact a trust region repairs—a result absent from the prior analysis. We report it in full, including the corner where it fires against us. By the sweep's own 5% rule, the tasks whose collapse survives tuning number $2/7$ with neither correction applied, $0/7$ under target scaling alone, $5/7$ under the protocol correction alone, and $4/7$ on the au-

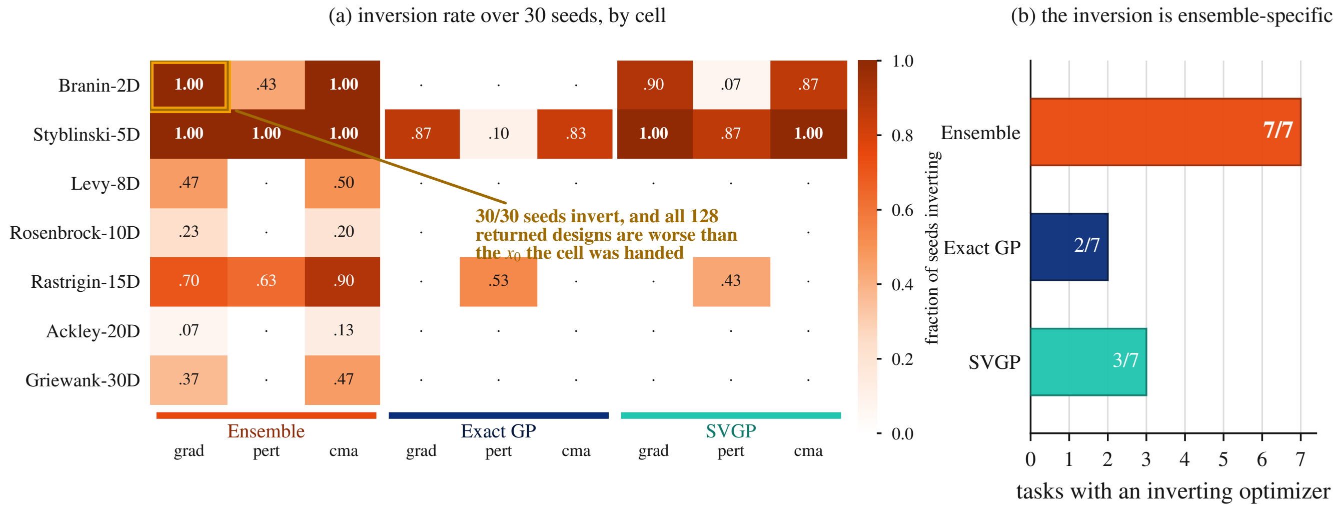

Figure 6: The ensemble's LCB ranks designs it invented above real designs it was handed. Each cell is seeded with the top-128 offline designs, which the candidate/oracle correction leaves in the returned pool; an inversion is a seed whose returned set is worse under the ground-truth oracle than that seed set. (a) Inversion rate over 30 seeds for all 63 cells—the ensemble block lights up and the exact GP's is almost entirely dark. On Branin-2D under gradient ascent all 30 seeds invert and every one of the 128 returned designs is worse than the best design the cell started with. (b) Counting tasks with at least one inverting optimizer: 7/7 for the ensemble, 3/7 for the SVGP, 2/7 for the exact GP. Audited engine, X1 and X3 on, K=5, β=2, 30 seeds.

dited engine. The collapse is genuine surrogate geometry on most tasks, driven by the candidate/oracle correction rather than target scaling—and on one corner not genuine at all.

Two corrections to the coverage analysis. First, the collapse is not gradient-specific. We registered that ensemble coverage would also be low on CMA proposals; it is. Under the matched protocol gradient and CMA proposals differ in out-of-distribution coverage by 0.010, against 0.117 under the unmatched one, because both pool every iterate and return the top 128 by surrogate LCB. The claim narrows to the ensemble paired with any aggressive selector, and we withdraw the gradient-specific framing by name. Second, the premise-coverage value of 0.97 in the prior analysis belongs to a scikit-learn exact GP, not the differentiable GP the grid scores, whose coverage averages 0.831 and falls to 0.38 on Styblinski-5D; both are reported with the model each belongs to. Coverage of $\{\mu - f \leq \beta\sigma\}$ is exactly the validity of the lower bound, which is what makes these frequencies worth measuring.

6 Design-Bench Results

Surrogate main effect and the omnibus. Across all four engine corners the Design-Bench surrogate main effect is essentially zero—η^2_{surr} of 0.001, 0.032, 0.002 and 0.018 on five tasks, every point estimate in the bottom tenth of its own bootstrap interval—and the Friedman omnibus fails to reject at p = 0.19 to 0.76, with η^2_{opt} at 0.050–0.173. We therefore report no detectable difference at this power, never equivalence: at n=7 a paired test needs $|d_z| \geq 1.27$ for 80% power at α=0.05—larger than anything we observe—and a failure to reject is not a demonstration that an effect is absent (Agarwal et al. 2021). Two robustness notes belong with the number. Dropping GFP leaves η^2_{surr} at 0.001–0.021 (η^2_{opt} 0.047–0.184) with the omnibus still not rejecting. Adding the two MuJoCo tasks raises it only to 0.044–0.091 but does make the omnibus reject in three of four corners (p = 0.011 to 0.014)—and the decomposition localizes what rejects. Side by side in those three corners, η^2_{surr} is 0.091, 0.044, 0.062 against an η^2_{opt} of 0.146, 0.197, 0.181: the surrogate effect stays at the floor of its own bootstrap interval on this MuJoCo-inclusive set exactly as it does on the five-task set—each of those three point estimates sits in the bottom fifth of its own interval, and the largest surrogate effect anywhere in the table (0.091) falls below the *smallest* optimizer effect in any rejecting corner (0.146)—while the optimizer effect runs 1.6 to 4.5 times larger. Those two intervals overlap inside every single corner (on_on: η^2_{surr} $[0.006, 0.305]$ against η^2_{opt} $[0.008, 0.468]$), so nothing here is a within-corner separation and we do not read it as one; the argument is carried entirely by the cross-corner ordering that follows, which needs no such separation. The cross-corner pattern settles it—the one corner that does *not* reject (p=0.16) carries the *larger* surrogate effect (0.053, against a rejecting corner's 0.044), so surrogate effect size does not track rejection, while η^2_{opt} tracks it cleanly (0.096 where the omnibus fails, 0.146–0.197 where it rejects). One reading of that rejection is that the null is *fragile*—that it holds only on a task set chosen to make it hold, and that adding tasks dissolves it. We name that reading because it is the strongest one available against this section, and the decomposition is what answers it: adding MuJoCo does move the omnibus, but it does *not* move η^2_{surr} out of the floor, raising it only from 0.001–0.032 to 0.044–0.091, still four to nine times below the synthetic grid's 0.405. A fragile surrogate null would show a surrogate effect that grows with the task set and tracks the omnibus; ours grows barely and tracks it inversely. What the added tasks recruit is the optimizer axis, which is why we report the rejection under its own decomposition rather than dropping the two tasks that produce it. We also disclose a pooling defect in the prior analysis: its normalization ranged over eleven cells, two of them arms the grid never defines. Because mean-rank conclusions depend on the pool (Benavoli, Corani, and Mangili 2016), we unify to the nine defined cells before any rank claim.

Matching the query budget strengthens the optimizer axis. Everything above is measured at native budgets, where gradient outspends perturbation by $6.25\times$ in the X3-off corners and $11.8\times$ in the X3-on ones, so the obvious objection is that the optimizer axis on Design-Bench is search intensity wearing a costume. We equalized surrogate queries across the full seven-task grid, both MuJoCo tasks included, at Q=25,600 (X3-off) and 51,456 (X3-on); gradient lands exactly on target, perturbation between 0.00% and -0.50%, and CMA $+0.05$ to $+0.09\%$, with none of the 252 cells more than 5% off. *The correction runs backwards to the objection.* Matching does not shrink the optimizer effect; it raises it by half again to nearly double in every corner—1.46 to $1.88\times$, η^2_{opt} of $0.096 \rightarrow 0.180$, $0.145 \rightarrow 0.255$, $0.193 \rightarrow 0.282$ and $0.181 \rightarrow 0.281$—while η^2_{surr} *falls* in all four, to 0.043, 0.062, 0.037 and 0.050. It also moves off_off from failing to reject (p=0.140) to rejecting (p=0.0179), so under matched budgets *all four* corners reject the omnibus where three did before, at $p = 0.0179$ to 0.00057, and perturbation leads the optimizer marginal in all twelve corner-by-level combinations. The native imbalance was *masking* the optimizer effect on Design-Bench, not manufacturing it: in this suite gradient's extra queries were buying it nothing that perturbation could not buy more of. Two limits are ours and neither is optional. First, this is *established at high budget*. At the matched low level the picture weakens in both directions—η^2_{surr} reaches 0.126 in one corner, above our pre-registered 0.10 floor, and in another the two axes are effectively tied (η^2_{opt} 0.085 against η^2_{surr} 0.081)—so the primary level decides, as pre-registered, and the phrasing mirrors the synthetic arm's own low-budget disagreement rather than hiding it. Second, the within-corner intervals overlap here as they do everywhere in this paper (on_on: η^2_{surr} $[0.004, 0.339]$ against η^2_{opt} $[0.137, 0.490]$), so what is licensed remains point-estimate localization across corners, never a within-corner separation. The "all four reject" count is the seven-task set; on the five-task and GFP-dropped sets three of four reject under matching, in the same direction and with a larger η^2_{opt}.

Two validity checks on that comparison. The matched runner fits each surrogate once per task and seed and reuses it across optimizers, where the main grid rebuilds per cell, which moves perturbation's position in the random stream. We therefore ran a native-budget control through the same runner rather than comparing against the published numbers directly. It reproduces the published corners to within 0.004 on every η^2 and puts every Friedman p on the same side of 0.05, so the native-versus-matched contrast is a budget contrast and nothing else—the effect is caused by the queries, not by call order. Separately, GFP is where this confound could most plausibly have manufactured the result: its CMA cell spends 570 surrogate queries against gradient's 51,456, a $90\times$ imbalance rather than the $8\times$ seen elsewhere, because at d=4,740 the library falls back to separable CMA and converges almost at once. Matching raises that cell to 51,494 queries, in line with every other. The seven-task and GFP-dropped sets move together, so repairing that starved cell did not by itself drive the outcome—but the sharpest available budget objection is now removed by measurement rather than argued away.

Robustness to oracle approximation. We pre-registered this competing mechanism and the test that would kill it. The null survives on the exact-oracle subset in every corner ($p = 0.34$ to 0.51), while within this oracle-subset analysis the only rejections are on the random-forest-oracle subset ($p = 0.021$ to 0.033 in three of four corners); the MuJoCo-inclusive and budget-matched rejections reported above are a different partition of the suite and are not in tension with this one. Repeated oracle evaluation gives a variance at machine precision ($\leq 2.3 \times 10^{-13}$, and 3.6×10^{-15} on four of five tasks), so the null is not noise drowning a signal. The counter-objection is ours to state: with two exact-oracle tasks this subset test has very little power, so the result is consistent with the null rather than a powered acceptance, corroborated by $\eta^2_{\text{surr}} \approx 0$ and the noise floor.

Frozen cells are a benchmark-construction finding. Part of this null is structural rather than statistical, and it is a reusable result in its own right: as constructed, several of Design-Bench's cells are degenerate for surrogate-attribution studies, freezing at the dataset reference so that they carry neither a main effect nor an interaction. On TF-Bind-8, the three exact-GP cells return exactly 1.0 on all 16 seeds with zero variance in every corner, and ensemble-with-perturbation joins them in two of the four—four frozen cells of nine there, three in the others. That value is not a score: scores are min–max normalized to the offline dataset's own range, whose maximum is attained by multiple tied rows already in the data, so a cell reporting 1.0 returned nothing better than its input. On Ant, two of three GP cells return the bit-identical constant 1.52874195557571411 with standard deviation exactly 0.0 across all sixteen seeds, and a third (SVGP with perturbation) hits it on ten of sixteen. Three cells across two surrogate classes converging on one bit-identical value is retrieval, not coincidence. The count is concrete—up to four frozen cells of nine on TF-Bind-8, at least two of three GP cells on Ant—so an attribution study on this benchmark must screen for frozen cells before it decomposes, or it reads a surrogate axis that cannot move as evidence about surrogates. This does not rescue our own positive signal, which stays synthetic-only; it identifies why the field's standard benchmark cannot answer the attribution question as posed.

The mechanism label is the opposite of the one we had. Decode snap-back—movement in a relaxed logit space that argmax-decodes back to the starting design—predicts that an exact-constant freeze requires a decode step to snap through. Ant is continuous with no argmax decode anywhere in its path, so our pre-registered kill against that account fires directly. The surviving mechanism is LCB paralysis: the GP's lower confidence bound is locally maximal at the data, so the optimizer never leaves—a freeze occurring even without a decode step. We rest that on our own evidence, which is the stronger half: sixteen bit-identical seeds across two GP implementations on a continuous task with no argmax decode. The closest prior formalization is an inconsistency result for expected-improvement-driven Bayesian optimization, which proves that a run started at the true optimum can have a trajectory that converges back to that point and is not dense in the domain (Yarotsky 2013). It differs from ours on three axes—an expected-improvement acquisition rather than a confidence bound, online rather than offline, an adversarial worst-case start rather than a seed-robust empirical observation—so we cite it as the nearest prior art and withdraw any suggestion that the failure mode is unowned, without claiming our observation is an instance of that theorem. The contrast with the trust-region literature is instructive, the vocabulary similar and the pathology opposite: there locality is a *deliberate* remedy for over-exploration in high dimension (Eriksson et al. 2019), whereas here it is an unintended terminal state the acquisition reaches on its own.

The Design-Bench inversion and the synthetic null are one finding. This section's most awkward number is that the optimizer axis *inverts* here—perturbation leads in all four corners (0.76–0.81 against gradient's 0.40–0.68) and η^2_{opt} exceeds η^2_{surr} *in point estimate* in every corner of every task set we report—the within-corner bootstrap intervals overlap here as they do everywhere in this paper, so this is a localization of point estimates and never a within-corner separation of the two axes—while in the synthetic grid of Section 4 the optimizer main effect is small (η^2_{opt}=0.038 $[0.003, 0.123]$ under matched budgets), a magnitude measured in that grid and never a verdict that optimizer choice is negligible in general (Section 7). Read through the frozen cells, the two suites stop disagreeing. If the GP cannot be moved, every optimizer scores it identically, the between-optimizer variance that survives comes almost entirely from the unfrozen ensemble cells, and the axis inverts as an arithmetic consequence of the freeze rather than as a substantive contradiction between benchmarks. A frozen surrogate reports the optimizer's variance because its own is gone—the payoff of a decomposition over a leaderboard. Four limits are ours. It is two of three Ant GP cells, not three—the gradient cell returns 1.3242 ± 0.1975 and is not frozen. Fact and label are separable, so losing the label does not lose the frozen cells. Frozen cells and genuinely near-tied cells are reported separately, never pooled. And the arithmetic reading is not the whole of it: the freeze explains why the axis inverts, not how large the optimizer effect is, and the magnitude is what the matched-budget arm establishes.

7 Discussion and Limitations

Pre-registered predictions and their outcomes. Four registered predictions came back against us and two registered mechanism arms failed to deliver what they were built for; we report all six, because undisclosed they would read as selective reporting to anyone who opens the repository. Our registered headline was that the acquisition optimizer explains most of the reported gap: in this grid it does not, η^2_{opt} being 0.038 even after correcting the budget imbalance that suppressed it, and the pre-committed contingency—pivot to the mechanism rather than quietly restate—is what this paper does. The ensemble-size reversal criterion did not fire, but its accompanying prediction confirmed: the headline is reported at the K that maximizes it (Confound 3). The β–σ criterion fired outright, narrowing Confound 4 to its per-task form, and the gradient-specific coverage framing was retracted on our own registered test. The two mechanism arms are the most expensive: smoothing the ensemble's mean was built to confirm the account we had published and refuted it, and the phantom-maxima probe was built to establish a mechanism and returned a seventh elimination.

Interpreting the optimizer effect. Under a matched surrogate-query budget of Q=51,456, $\eta^2_{\text{opt}} = 0.038$ $[0.003, 0.123]$, so optimizer choice explains little of the variance in this grid. Three qualifications travel with it. It replaces our own unmatched 0.005 that understates by roughly eightfold, a correction we disclose. The confirmation is not comfortable: the upper bound of 0.123 lies inside our preregistered inconclusive band, so an effect up to about 0.12 is not excluded at n=7—the same precision limit that makes the corner decomposition unresolvable. And the secondary budget level does not corroborate cleanly (0.066 $[0.014, 0.340]$ at Q=4,352), so the null is established at high budget and underpowered at low, never unqualified. Most importantly, the *best* optimizer changes with budget: gradient leads at low budget (0.731) and perturbation at high (0.732), where the unmatched grid showed gradient modestly best throughout—an $11.8\times$ imbalance hiding itself by making one optimizer look mildly best everywhere. As Section 4 measures, the small *marginal* is not a verdict that optimizer choice is arbitrary: its *conditional* effect through the interaction is four to thirty-three times larger, so which optimizer to use depends on the surrogate—the one thing a factorial and not a one-factor-at-a-time design can say.

Two confounds whose removal strengthens what they were meant to explain. The synthetic and Design-Bench budget arms now make the same shape of finding on two independent axes. Correcting the protocol confounds raised the surrogate main effect on the synthetic grid from our decomposition of PGS's Table 1 ($\eta^2_{\text{surr}} \approx 0.37$) (Chemingui et al. 2024) to 0.405 $[0.290, 0.556]$; equalizing query budgets on Design-Bench raised the optimizer main effect by half again to nearly double in every corner rather than dissolving it. In both cases the confound was suppressing the effect it was suspected of manufacturing, and in both cases the direction runs against the reality-check genre's, where corrected effects shrink (Melis, Dyer, and Blunsom 2018). We state the limit of that pattern as plainly as the pattern. It is two instances, not a law: the ensemble-size confound moves the surrogate effect the other way (Confound 3), so we claim neither that de-confounding generally strengthens nor that these two corrections share a mechanism. What they share is that a reviewer's prior about which way a control will move is not evidence about which way it moved, and both of ours were measured rather than assumed. The pattern survives bias correction and grows slightly under it: η^2 is positively biased at small n, and our own bootstrap artifacts put that bias at $+0.0099$ to $+0.0184$ across the four corners. Bias-correcting the two-confound comparison gives $0.351 \to 0.395$, an increment of $+0.0437$ against the $+0.0376$ of the uncorrected figures. We report ϵ^2 rather than ω^2 as the correction, on simulation evidence naming ϵ^2 the least-biased of the standard indices rather than the folk preference for ω^2 (Okada 2013).

No landscape covariate we tested predicts the gap. A natural objection is that the surrogate gap is really a landscape property in disguise. We tested every covariate we can construct and none predicts it at this power. The gap's rank correlation with dimension is $+0.107$ under gradient, $+0.036$ under perturbation and $+0.071$ under CMA—indistinguishable from zero under every optimizer ($p \geq 0.81$), and pointing the opposite way to the qualitative expectation that GP advantage should decay with dimension as neural surrogates learn non-stationarity that GPs need human intervention to capture (Li, Rudner, and Wilson 2024). That is a measured refinement of a named prior expectation, not a disagreement about anything either of us tested directly. Textbook multimodality actively mis-sorts: within the "many local minima" family the gap spans three orders of magnitude, while the one strictly unimodal task has among the smallest gaps. Separability fares no better. This is a null from looking rather than from failing to look, and it is not an artifact of a flattening reparametrization—each task is mapped from the shared unit cube back onto its exact native textbook domain.

One structural statement does survive, and we weight it honestly. The per-task gap *ordering* is stable across optimizers (Spearman 0.71–0.96; the gradient-versus-perturbation pair at 0.71 does not reach significance at n=7, p=0.07) while the gap *magnitude* collapses under perturbation. Those are not in tension—one is relative ordering across tasks, the other absolute magnitude within an optimizer—and together they constrain any surviving mechanism to be multiplicative, a task factor scaled by an optimizer-aggressiveness factor. The collapse is bootstrapped; the ordering stability is descriptive at n=7, so the collapse is the better-evidenced half and the multiplicative reading is a constraint on future accounts rather than an account.

What the null does and does not license. We never claim equivalence on Design-Bench, and we can say more than that disclaimer. The released code computes a paired two-one-sided-tests bound, which we report rather than withhold: the best-versus-worst gap is 0.376 with a 90% interval of $[-0.108, 0.860]$ and an effect bound of 0.484, which is *not* equivalent at either a 0.5 or a 0.3 margin (Lakens 2017). So the null is a failure to detect that also fails to bound the effect below a margin anyone would call negligible, and rejecting small effects would require samples we do not have. That is a positive statement about the null's reach and it is weaker than equivalence in the direction that matters.

Robustness of the decomposition to its normalizer. Because the per-task normalizer is an operating-point coordinate rather than a neutral choice, we recomputed the decomposition under z-score and rank transformations in every corner (Table 3), and under leave-one-task-out. Every combination preserves $\eta^2_{\text{surr}} > \eta^2_{\text{opt}}$; magnitudes move and the interaction roughly halves under rank, but the direction does not. Leave-one-task-out moves the headline between 0.373 and 0.485, and we correct one expectation of our own: Griewank-30D, whose raw score range is by far the widest and which we had flagged as the likely dominant lever, is the *smallest* one—dropping it moves 0.405 to 0.395, while dropping Styblinski-5D moves it to 0.485.

Scope of the optimizer-axis claim. We make no field-reversal claim. That the choice of statistical model often matters more than the acquisition heuristic is the organizing doctrine of the standard Bayesian-optimization review (Shahriari et al. 2016) and is a decade old; it governs online BO and the choice of acquisition *function family*, not the numerical search routine in the offline setting. What our design falsifies is the local premise of one recent paper—that offline black-box optimization has focused on improving surrogate models while using fixed search strategies (Chemingui et al. 2024)—by holding one shared protocol and varying both axes. We make no claim about the field's general beliefs about surrogates versus acquisitions. Nor do we claim ensemble size as a discovered mechanism (Li, Rudner, and Wilson 2024; Abe et al. 2022), equivalence on Design-Bench (Agarwal et al. 2021; Lakens 2017), priority on the K range (Li, Rudner, and Wilson 2024), or distance-aware uncertainty as ours (Liu et al. 2020; van Amersfoort et al. 2020).

Limitations. Seven synthetic tasks is a small n, binding the corner decomposition, the optimizer interval and the Design-Bench omnibus alike. The synthetic datasets are drawn once at seed 0, so reported variance is training and optimization variance, not data-draw variance. GFP is quarantined from coverage claims because its decode is degenerate, and we report that both ways: excluding it, one claim flips—the exact GP's mean in-distribution coverage moves from below nominal to above it—while the effect-size null is unchanged. Budget matching now covers both suites, which is what lets Section 6 assert its optimizer result rather than qualify it; what remains conditional there is the budget *level*, since both the optimizer effect and the surrogate null weaken at the matched low budget. Two of the seven controls are narrower than the count suggests: capacity is eliminated at fixed depth 2, so the depth-driven form of that objection is open, and σ is eliminated at one ensemble construction, so a likelihood-trained variant is untested. What the eliminations leave open is finite and namable—those two arms, plus whether the surrogate-geometry account and the optimizer interaction are one mechanism or two (Section 5)—so the argument from elimination bounds the residual rather than ranging over an open space. And the surviving mechanism statement is a diagnosis with seven eliminations behind it and no positive causal test in front of it. The one positive account we pre-registered—off-support phantom maxima under prior reversion—was refuted on its first limb and untestable on its second, because at this operating point the GP is nowhere in a reversion regime inside the feasible set. A second pre-registered attempt at a positive mechanism also returned an elimination rather than an account: we instrumented the far field to test whether the ensemble's mean grows without bound along rays outside the training support while the GP's reverts. The ensemble does extrapolate near-linearly on all seven tasks, but neither GP class reverts anywhere in the probed range—no ray curve of any class flattened out to three times the box exit radius, and on the highest-dimensional task the GP's far-field slope is the steeper of the two—so far-field functional form does not distinguish the classes and cannot carry the mechanism. Section 5 therefore ships as seven eliminations and a diagnosis, which is what the pre-registered binary licenses.

8 Conclusion

The reported advantage of GP-LCB in offline MBO is measured through five confounds specific to this comparison, and removing the two protocol confounds moves the surrogate main effect from our decomposition of PGS's Table 1 ($\eta^2_{\text{surr}} \approx 0.37$) (Chemingui et al. 2024) to 0.405 $[0.290, 0.556]$—against the direction each named audit in this genre ran, with no precedent we could find, and the net of two corrections against a third. That decomposition is not resolvable at seven tasks and we say so; the firmer results are that any $\eta^2_{\text{surrogate}}$ here is a joint artifact of five operating-point coordinates papers do not state, and that the surrogate$\times$optimizer interaction—the term this design exists to estimate and a one-factor-at-a-time study cannot—exceeds the optimizer main effect in every corner, with an interval excluding zero in all four. What the advantage *is* remains open, narrowed by seven controls: not σ, width, budget, held-out accuracy, mean roughness, premise coverage, or distance-to-support with surrogate inflation. The loss itself is measured—the ensemble's optima are worse in true oracle value at matched distance and matched inflation. On the optimizer axis the claim is narrow and stays narrow: under matched budgets $\eta^2_{\text{opt}}{=}0.038$ $[0.003, 0.123]$ *in this grid*, which falsifies the local premise of one recent paper—that offline black-box optimization improves surrogate models while holding search strategies fixed (Chemingui et al. 2024)—and claims nothing about the field's general beliefs, which the standard Bayesian-optimization review settled a decade ago (Shahriari et al. 2016). What we offer is the taxonomy, the protocol, and the engine stamp that makes either reproducible.

Ethical Statement

This work studies optimization methodology on public benchmarks and introduces no new data or deployed system.

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